

B19 Series B

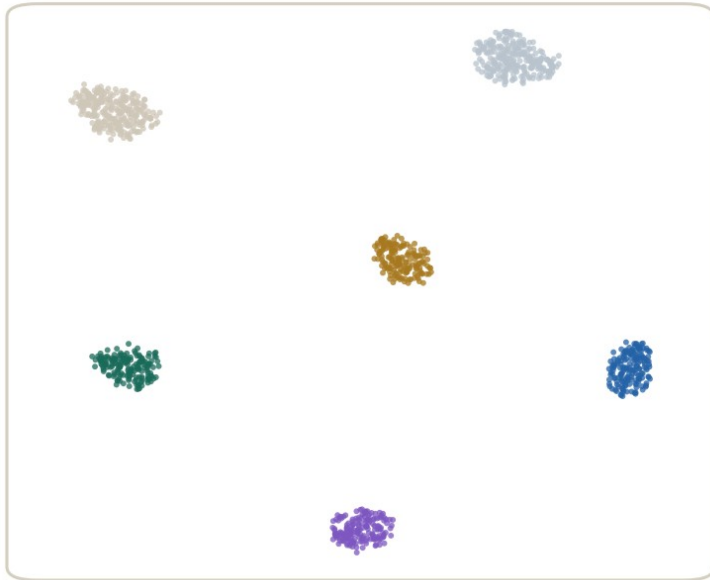
# Strategy design for complex samples: different tumours, different playbooks

Five decision points, working back from the biology — GBM, melanoma, breast, liver, AML, multi-region: six playbooks.

About 40 min | Prerequisites: B11, B12, B16 | Suggested order: B17 → B19 → B18

# One pipeline, three datasets, three conclusions

GBM, four patients



colour = malignant; grey = TME

**"four novel  
tumour subtypes!"**

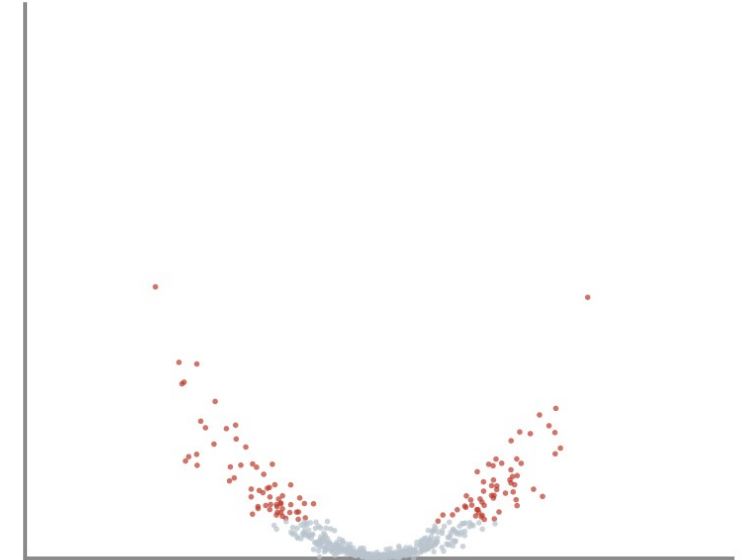
AML bone marrow



red = AML blasts; grey = normal marrow

**"clean CNV, no patient islands  
— no malignant cells found"**

melanoma, pre vs post



cell-level DE (actually run)

**"treatment changed 120 genes!"  
(true effects: 0)**

simulated data

**one PBMC tutorial pipeline, three datasets, three confident wrong conclusions**

All three conclusions are confident; all three are wrong — and no error was thrown

## WHERE THIS EPISODE STARTS

**The pipeline was not wrong.  
The strategy was.**

Every line of code ran correctly. The failure happened before the run — nobody asked what this data is and what the question demands.

# What you will learn

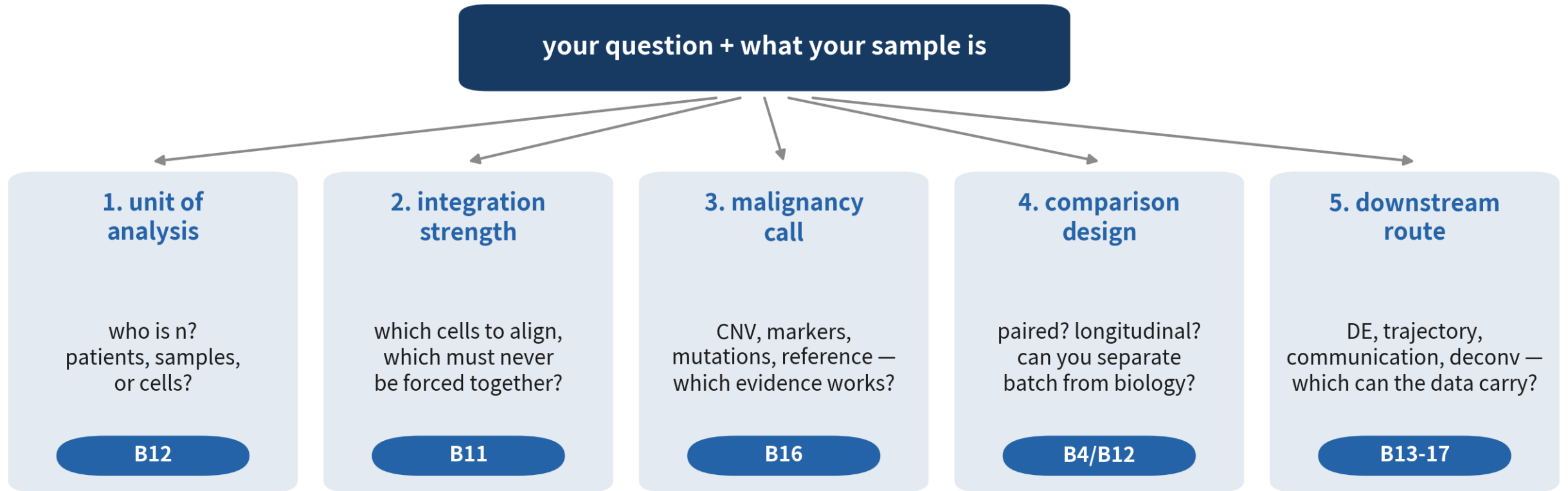
- 1 Design the analysis strategy for a complex sample before touching code, using five decision points
- 2 Explain how and why the playbooks differ across GBM, melanoma, breast, liver, AML, and multi-region designs
- 3 Read the strategy decision tree and pick a starting point for your own tumour data

# 01

## The five decision points

Work back from the biological question — never forward from the pipeline

# Five decision points



**answer the five decisions and the tools pick themselves — never the other way round**

Each point has its own episode for the mechanics; this episode wires them into one strategy

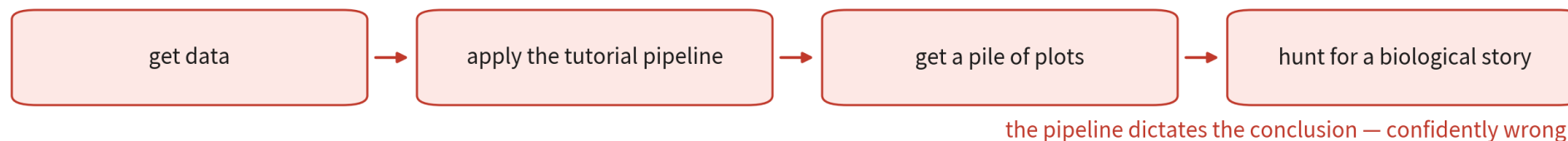
# Five decisions, five questions

These five questions are exactly what `strategy_checklist()` in `R/B19_strategy.R` prints

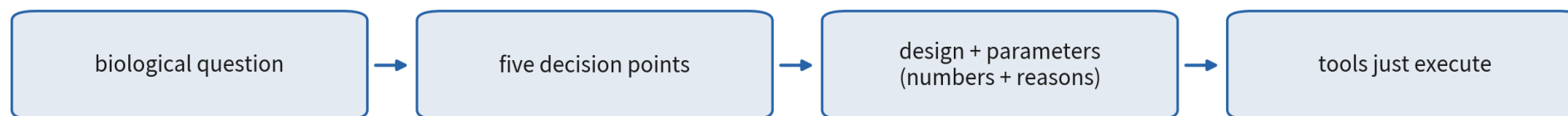
| Decision                | The question                           | Cost of a wrong answer                                |
|-------------------------|----------------------------------------|-------------------------------------------------------|
| 1. Unit of analysis     | Who is the independent observation?    | Pseudoreplication: thousands of false positives (B12) |
| 2. Integration strength | Which cells align, which must not?     | Patient biology erased as batch (B11)                 |
| 3. Malignancy call      | Which evidence exists for this cancer? | Wrong malignant/normal line poisons everything (B16)  |
| 4. Comparison design    | Can batch and biology separate?        | The 'treatment effect' is a batch effect              |
| 5. Downstream route     | Can the data carry this analysis?      | Forcing analyses the cells can't support              |

# Question-first, not pipeline-first

## pipeline-first (tutorial inertia)



## question-first (this episode's method)



same tools, same functions — the only difference is whether decisions happen before or after the run

- Pipeline-first: defaults decide your conclusion, confidently
- Question-first: answer the five decisions and the tools fall into place
- Both roads call the same functions — only the timing of decisions differs



# Strategy as code: strategy\_checklist()

```
source("R/B19_strategy.R") # 本集附的策略模板  
strategy_checklist()      # 動手前，五個問題過一遍
```

- B19 策略檢查表（動手前逐題作答並記錄） —
  - [1] 統計單位：這個比較裡，獨立的觀察值是誰？
  - [2] 整合強度：哪些細胞該對齊、哪些不能硬拉？
  - [3] 惡性判定：這種腫瘤，哪一種證據拿得到？
  - [4] 比較設計：批次跟生物效應拆得開嗎？配對了嗎？
  - [5] 下游路線：資料撐得起這條下游嗎？先驗最小格。

Write the answers into your analysis notes — numbers, reasons, records, like any QC threshold

## CORE IDEA

**Every parameter is a hypothesis:  
a number, a reason, a record.**

B5 said this about QC cutoffs. This episode promotes it to a principle for the whole pipeline.

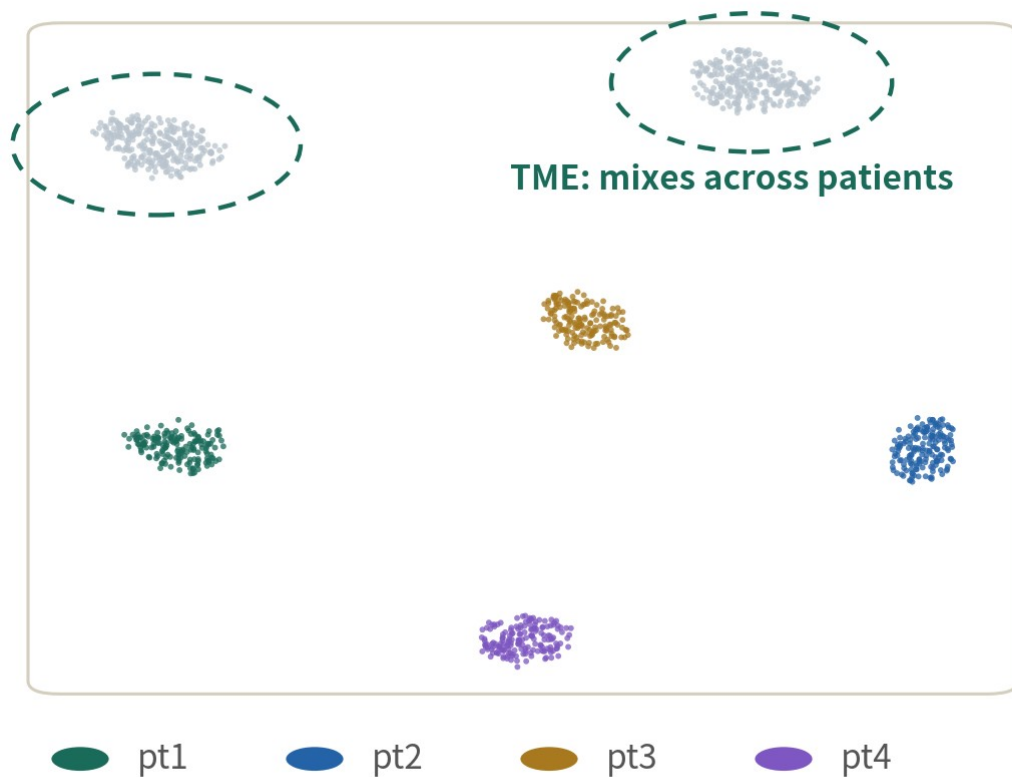
# 02

## GBM: the solid-tumour prototype

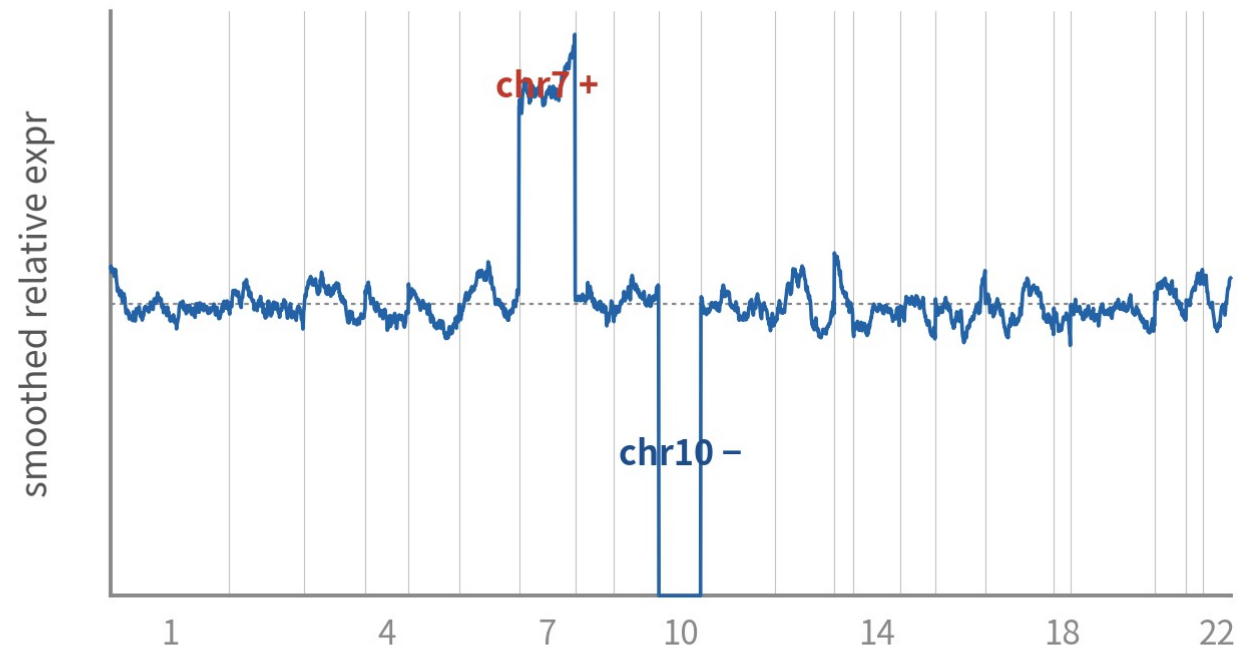
Patient islands, big CNV, four states, core/margin — the assembly of everything so far

# GBM's two landmarks

malignant: one island per patient



CNV large and canonical: chr7 gain, chr10 loss

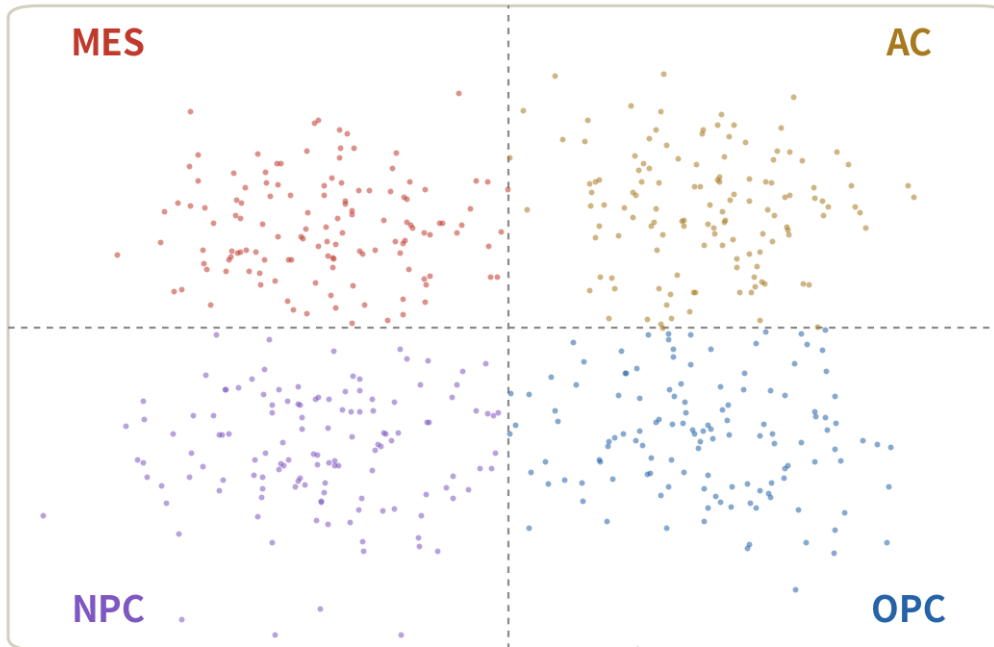


over 80% of GBM patients carry this pair — the anchor for malignancy calls  
simulated data

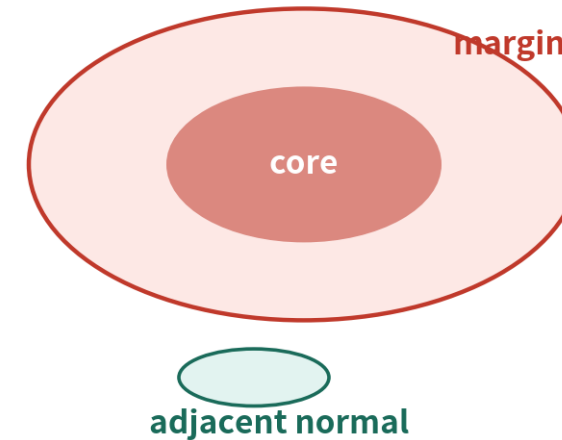
Malignant patient islands (B11's negative control) + chr7 gain / chr10 loss (B16's anchor) — the strategy grows from these two facts

# Comparing within malignant + how to sample

inside malignant: compare state scores, not clusters



sampling design: paired core / margin



## pair within patient

core vs margin absorbs patient effects — power even at  $n = 4$

## hypoxia gradient

necrotic core skews MES-like — composition itself is a result

## margin's value

few infiltrating cells, weak CNV — check MIN\_CELLS first

Neftel's four states: every patient has all four — scores compare across patients

schematic

Across patients, compare Neftel state scores, not clusters; sample core/margin pairs within patients

# GBM: the five answers

This is the strategy-card format — every tumour gets one

| Decision    | GBM's answer                                                             |
|-------------|--------------------------------------------------------------------------|
| Unit        | The patient; state fractions and scores use patients as n                |
| Integration | Align the TME for annotation; malignant cells per patient, unintegrated  |
| Malignancy  | inferCNV first (chr7+/10-), triangulated                                 |
| Design      | Paired core vs margin; check margin cell counts first                    |
| Downstream  | State scores; exclude hypoxia/stress signatures first; TME communication |

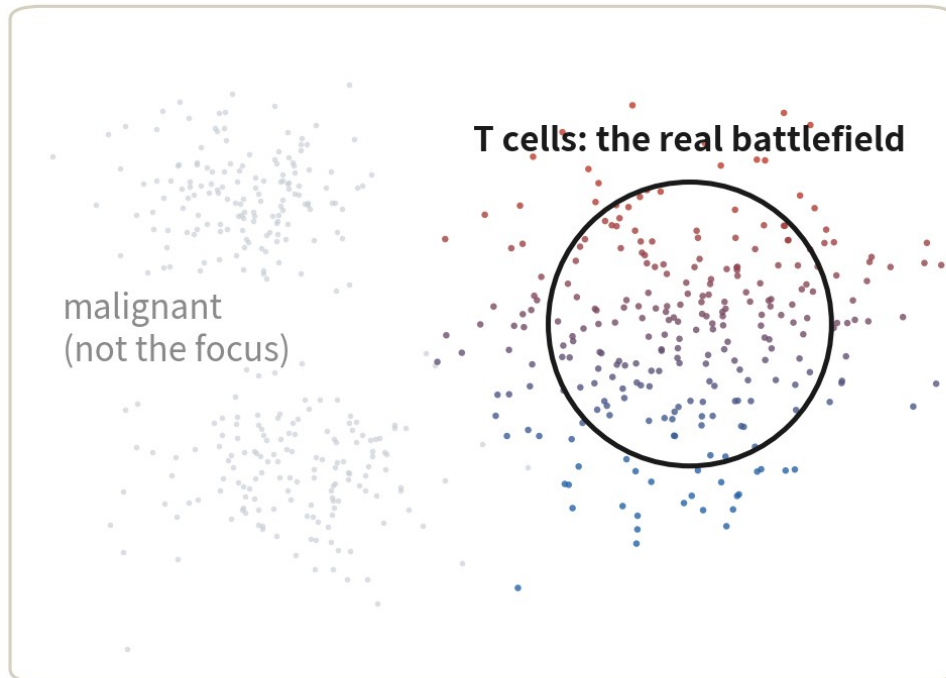
# 03

## Melanoma: the question isn't in the tumour cells

The immuno-oncology scenario — turn the lens to T cells

# A new protagonist: T cells and the exhaustion axis

immuno-oncology: aim the lens at the TME



exhaustion is an axis, not a switch

naive / memory

effector

exhausted

TCF7 · IL7R · CCR7

PDCD1 · TOX · LAG3

**subcluster deep enough**

extract T cells, redo HVG/PCA/clustering — the big UMAP hides them

**TCR is the trump card**

same clonotype = same family: track expansion and state shifts

schematic

Malignant calls still happen, but the research question lives in the TME — the strategy's centre of gravity flips



# Responder vs non-responder: the design

Immunotherapy's most common comparison — both landmines are design-time

## the disaster

### Cells as n, composition ignored

- 100k cells per arm — absurdly small p-values
- composition shifts misread as expression changes
- response defined post hoc, groups cherry-picked

## the sound one

### Patients as n; composition and expression asked separately

- per-type pseudobulk with patients as samples (B12)
- composition via proportions, expression via DE — two questions
- response criteria fixed before enrolment

# Melanoma's three cards

## Subcluster depth



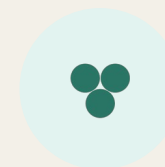
Extract T cells and redo HVG/PCA/clustering — the global UMAP hides exhaustion states

## TCR information



5' chemistry + TCR-seq: track clonal expansion and state shifts by clonotype (B4's multimodal)

## Easier malignant calls



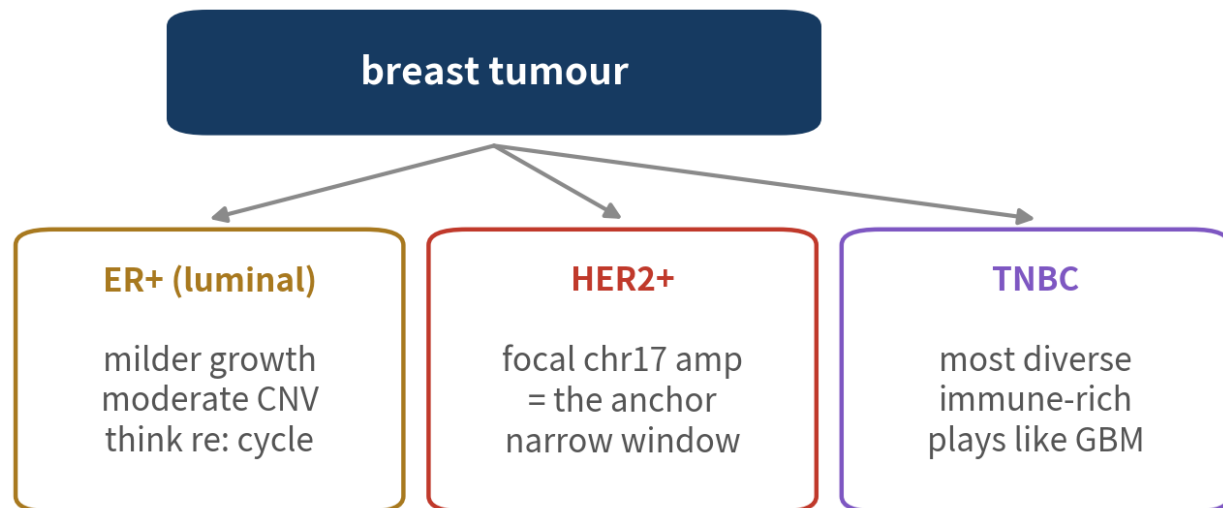
Melanocyte markers (MLANA, PMEL) plus CNV support usually suffice

# 04

## Breast cancer: subtype before everything

ER / HER2 / TNBC are not annotations — they are three diseases

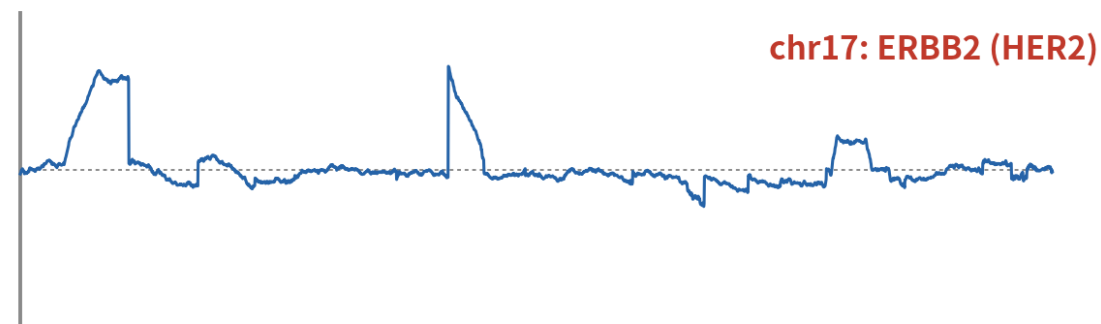
# Subtype triage + the focal HER2 amp



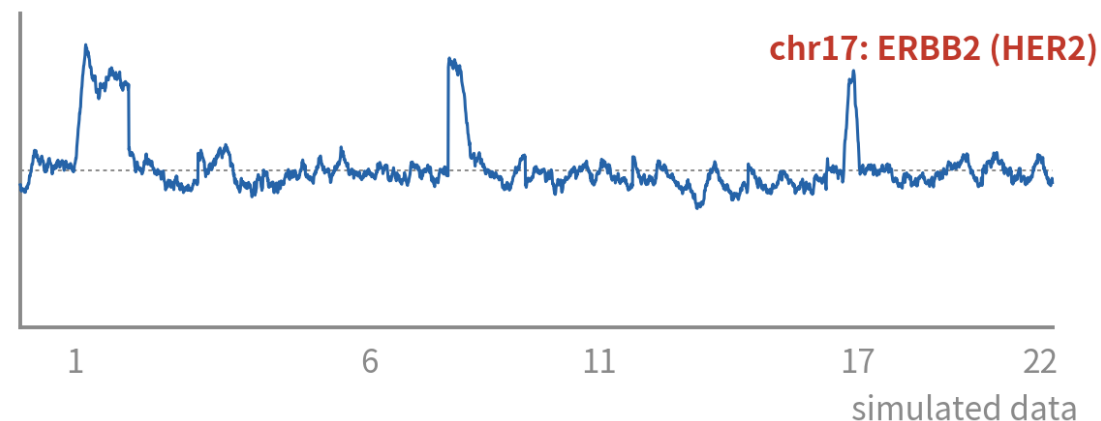
**subtype comes first: different subtype, different disease**

**the value of paired normal breast**  
normal epithelium = CNV reference + control —  
often available from mastectomy; plan for it

**wide window (101): the HER2 spike is smoothed away**



**narrow window (31): the focal amp appears**



A focal amp is a small event: inferCNV's default 101-gene window smooths it away — window follows event size

# Breast cancer's three decisions

## Triage, then analyse



Mixing subtypes turns DE into subtype composition — stratify at enrolment

## Narrow window for focal amps



Focal events like HER2 need window\_length near 31; the cost is more noise

## Paired normal epithelium



Normal breast = CNV reference + control; in epithelial tumours EPCAM+ is nearly tumour

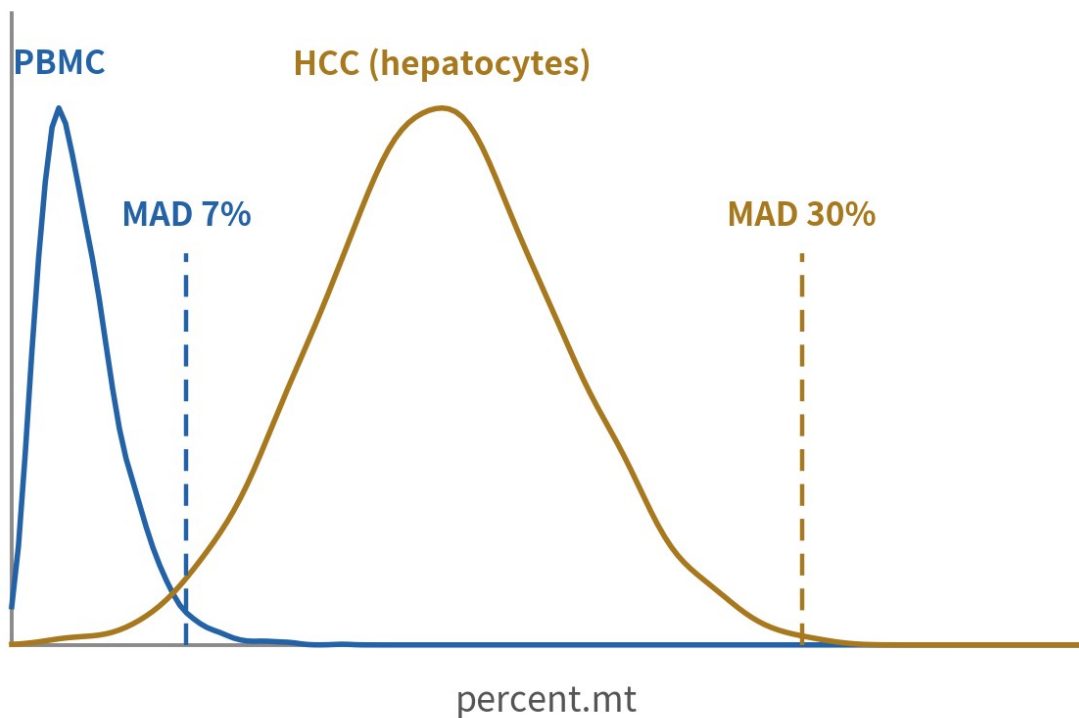
# 05

## Liver cancer: metabolism and virus

Reset the QC baselines, add the virus to the reference, learn zonation

# HCC's three special decisions

hepatocytes run high-mito: the whole baseline shifts



same MAD rule, entirely different numbers

reference decision: add HBV to the genome

GRCh38 + HBV

→ viral transcripts counted;  
HBV+ malignant cells surface

hepatocyte zonation is a continuous gradient

periportal  
(gluconeogenesis)

pericentral  
(detox / CYP)

normal zonation masquerades as 'subclusters' —  
learn it before calling malignant states

simulated data

The shifted mito baseline is tissue character, not death (B5's strategy room, full version); HBV in the reference is B2's mkref for real

# The HCC strategy card

## Re-derive all QC



Median mito can hit 15–20%; a copied 5% wipes out hepatocytes — share the rule (MAD), recompute the numbers

## Reference decision



Add HBV/HCV to the genome before mkref: viral transcripts become countable and flag malignant cells

## Zonation is not subclusters



The portal-central gradient gets sliced into fake subclusters; fibrotic CAFs and immune cells are their own storyline



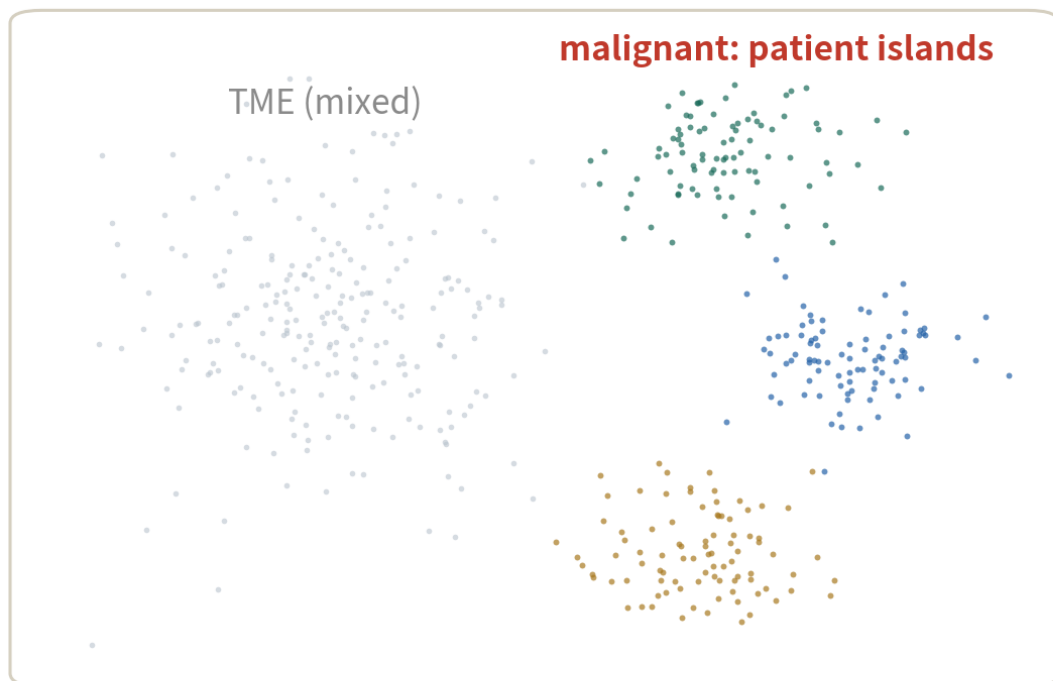
# 06

## AML: nearly everything reverses

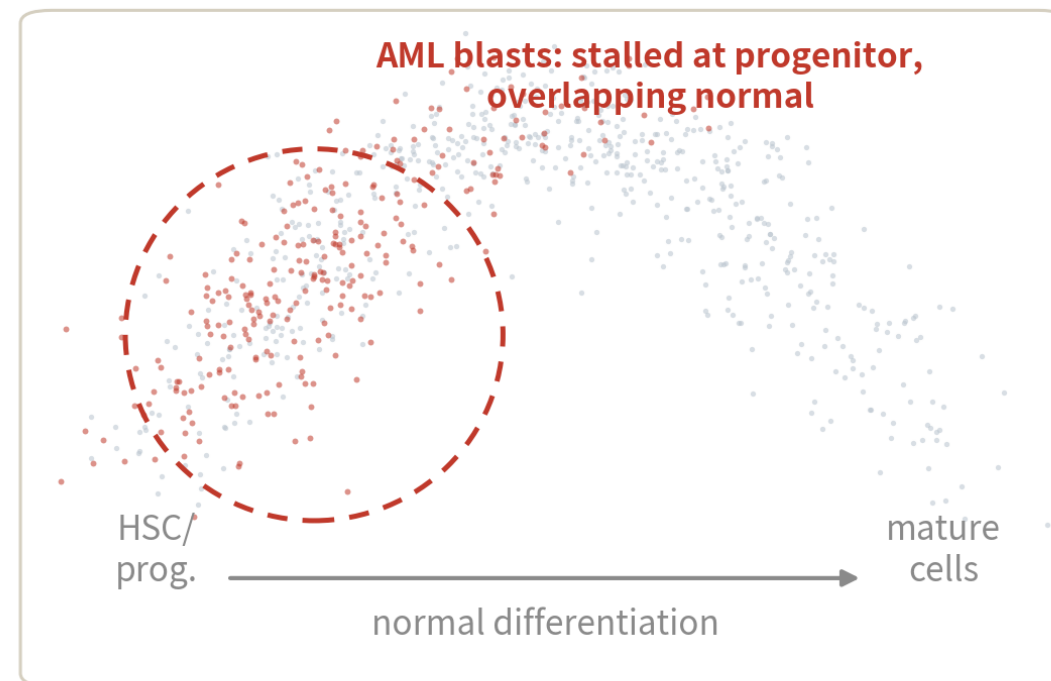
Blood cancers are the hardest class — all three solid-tumour signals fail

# Malignant cells that mimic normal

solid tumour: malignant cells stand apart



AML: malignant cells hide — the block is the signal



schematic

**all three solid-tumour signals (patient islands, CNV, markers) fail in AML**

AML blasts stall at the progenitor stage and overlap normal progenitors — the stand-apart signal becomes hide-inside

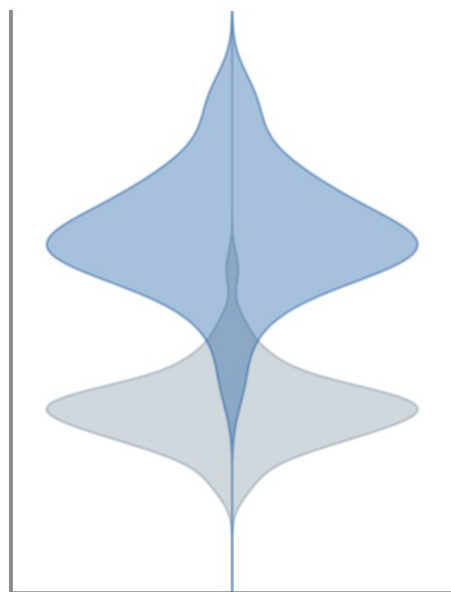
# Why the CNV route fails here

malignant vs normal CNV score: smaller events, less separation (grey = normal)

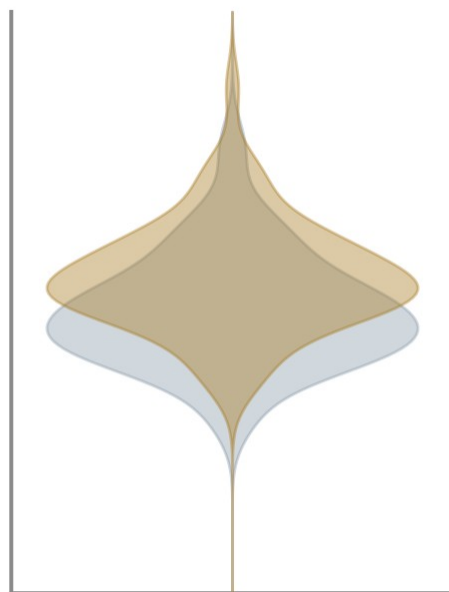
GBM-like: two whole arms

one small event

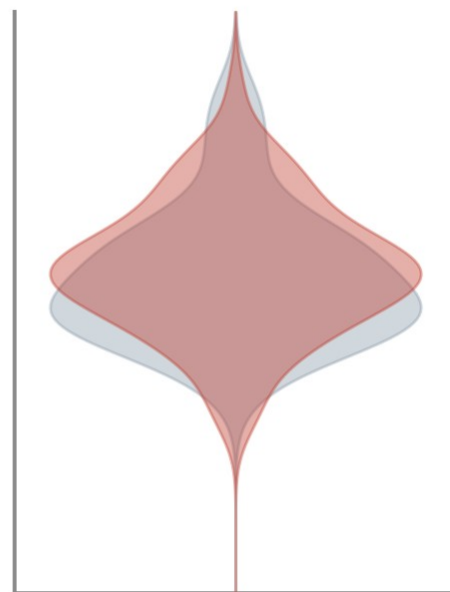
near-diploid (typical AML)



AUC 0.99



AUC 0.69



AUC 0.56

same inferCNV-style algorithm each time — the tool didn't change, the signal isn't there

The same algorithm on three event strengths: AUC falls from 0.99 to 0.56 — near-diploid scores simply don't separate

when scores overlap, switch evidence:

mutations / fusions  
(defining variants)

immunophenotype  
(surface antigens)

differentiation block  
(vs normal reference)

**CNV is one line of evidence, not a law**

simulated data

# Solid tumour vs AML: the reversal

Same questions, nearly opposite answers — the extreme case of different playbooks

solid (GBM-type)

## Malignant = the anomaly

- patient islands are a signal
- big canonical CNV — inferCNV leads
- integration's negative control = malignant stays apart

AML

## Malignant = deviation from a reference

- blasts blend into the normal continuum — islands vanish
- CNV often clean: mutations, fusions, immunophenotype
- a normal-marrow reference atlas is mandatory, not optional

# AML: the five answers

Read side by side with GBM's card to see how complete the reversal is

| Decision    | AML's answer                                                                                            |
|-------------|---------------------------------------------------------------------------------------------------------|
| Unit        | The patient — the one answer that never flips                                                           |
| Integration | Co-embed with a normal-marrow reference: disease = the deviation                                        |
| Malignancy  | Differentiation block + mutations/fusions + phenotype; CNV is auxiliary                                 |
| Design      | Longitudinal diagnosis/remission/relapse pairing is the core design                                     |
| Downstream  | Composition along the hierarchy, stemness scores; trajectories are meaningful (a real hierarchy exists) |

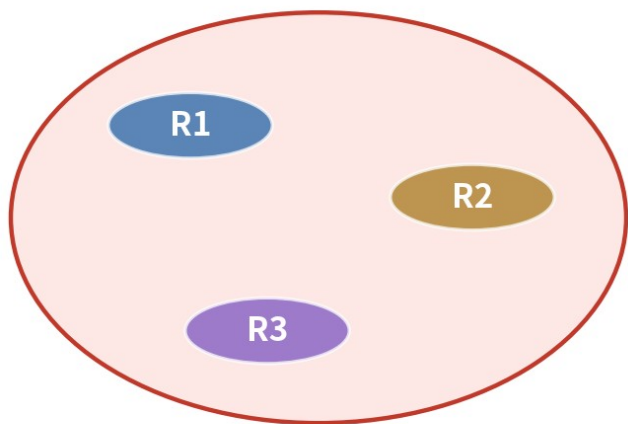
# 07

## Multi-region and longitudinal designs

Multiple samples from one patient — the rules follow 'within patient'

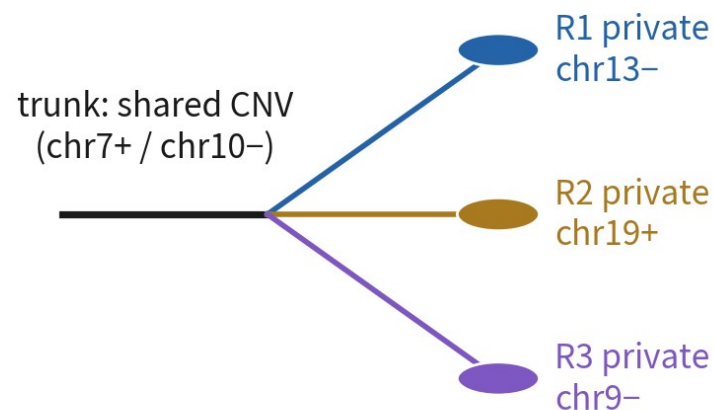
# Reading subclone evolution

one tumour, three regions



regional differences =  
space × evolution × microenvironment

subclone phylogeny



**trunk events = early**

CNV shared by all regions —  
candidate initiating events

**branch events = late**

private to one region —  
local evolution

**integrate within patient**

regions share the trunk — pool them;  
keep patients separate

schematic

Trunk = early, branches = late; pool regions within a patient, keep patients apart (B16's subclone reading becomes a design rule)

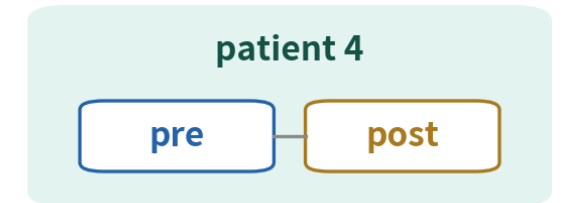
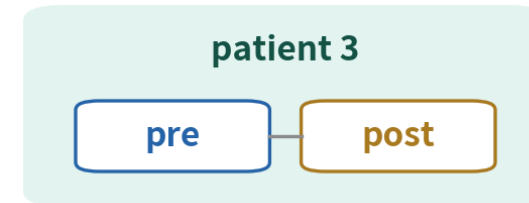
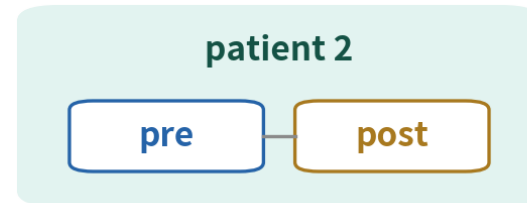
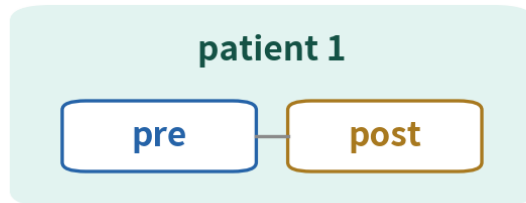
# Longitudinal: the batch disaster and its fix

the disaster design: pre/post = two batches



"treatment effect" = batch effect — mathematically inseparable

the design-time fix: pairing + mixed batches



pair pre/post within patients; bank samples and mix conditions within batches — only then can batch and biology separate

If pre/post equals batch 1/2, the effects are mathematically inseparable — only design can fix this (B4 and B11 foreshadowed it)



# Three things to do at design time

## Bank and batch



Freeze samples (nuclei workflow) and process pre/post pairs in the same batch — only then do batch and condition separate

## Count every cell first



The smallest patient  $\times$  site  $\times$  type cell decides what you can ask; estimate MIN\_CELLS before enrolment (B12)

## Pairing first



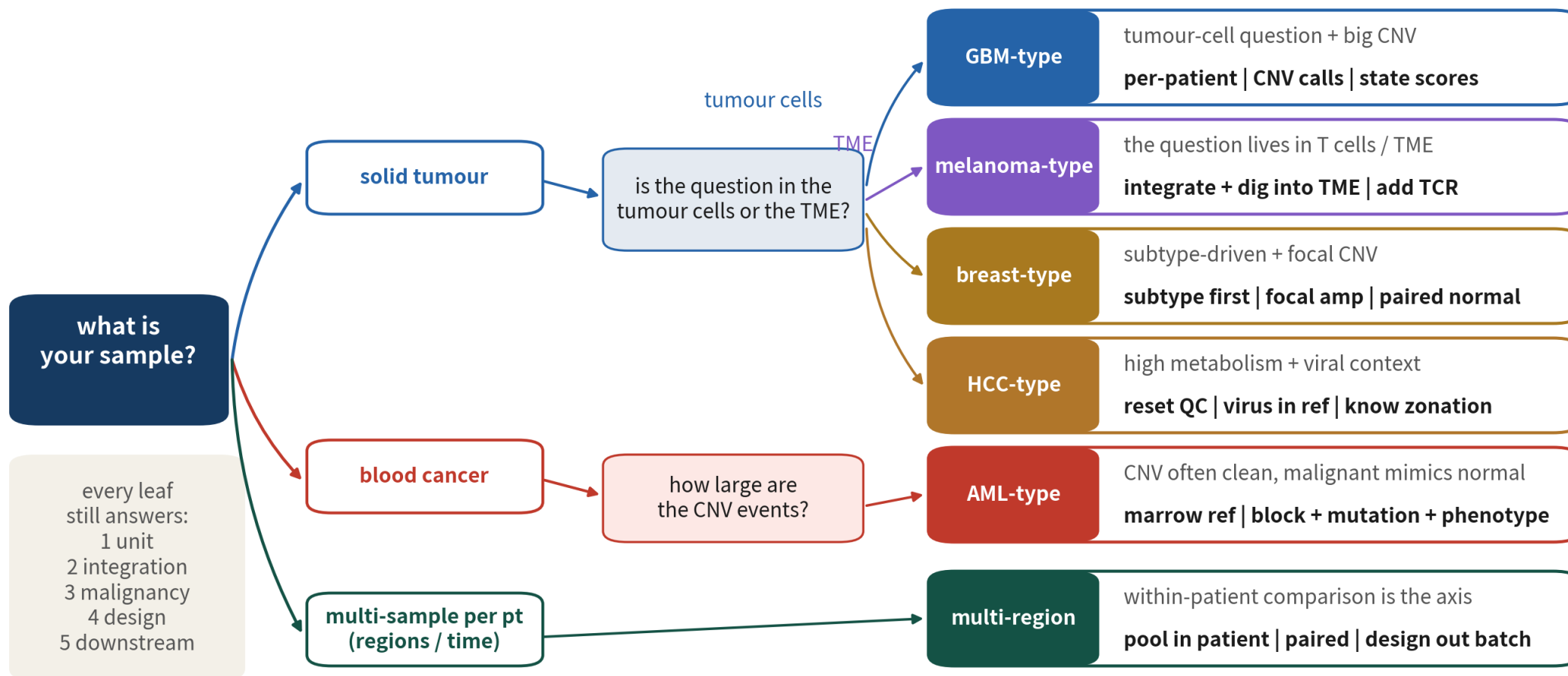
Within-patient pairing absorbs patient effects — power even at  $n = 4$ ; the one statistical bonus small studies get

# 08

## Six playbooks, one picture

The signature figure: from 'what is your sample' to a full strategy

# The strategy decision tree



six leaves are starting points, not dogma — your tumour may straddle two; walk the five decisions and you won't get lost

Print it and stick it next to your screen. The six leaves are starting points, not dogma — walk the decisions and you won't get lost

# Six scenarios, one line each

Each line compresses its section — if you can justify it, you've got the episode

| Scenario                    | One-line playbook                                                     |
|-----------------------------|-----------------------------------------------------------------------|
| GBM-type solid              | Malignant per patient; CNV calls; compare state scores                |
| Melanoma-type               | Aim at the TME: dig into T cells, add TCR, tumour takes the back seat |
| Breast-type                 | Triage by subtype; narrow window for focal amps; paired normal        |
| HCC-type                    | Reset QC baselines; virus into the reference; know zonation           |
| AML                         | Normal-marrow reference; block + mutation + phenotype for malignancy  |
| Multi-region / longitudinal | Pool within patient; paired tests; design out batch                   |

# 09

## Where people get hurt

Three traps — each with consequences actually computed

# Trap 1: the PBMC pipeline, applied verbatim

## What it is

Feeding tumour, AML, or pre/post data through the tutorial pipeline unchanged.



## Why it happens

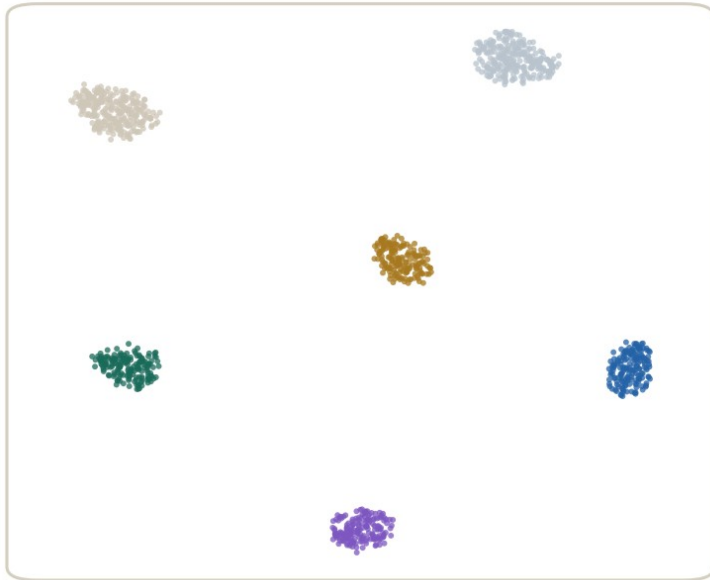
Every step completes and every plot looks fine — nothing signals that the strategy is wrong.

## What it costs you

The cold open's three conclusions: islands become 'subtypes', AML becomes 'no malignancy', batch becomes 'treatment effect'.

# Anatomy of three wrong conclusions

GBM, four patients



colour = malignant; grey = TME

**"four novel  
tumour subtypes!"**

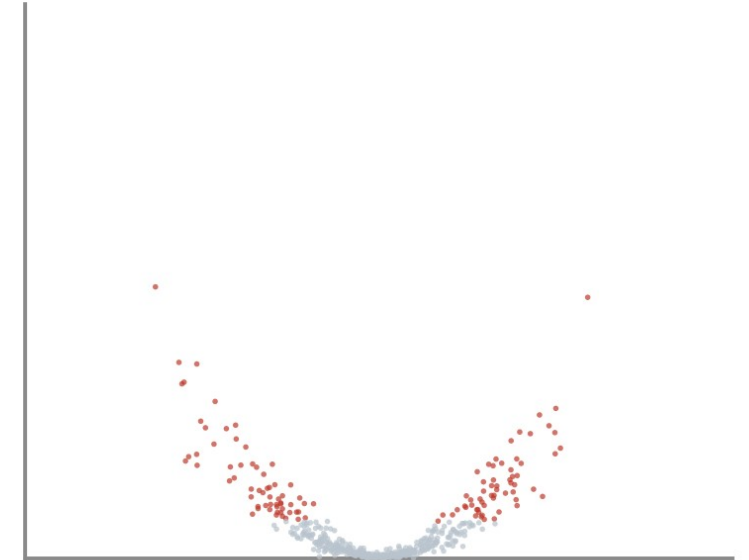
AML bone marrow



red = AML blasts; grey = normal marrow

**"clean CNV, no patient islands  
— no malignant cells found"**

melanoma, pre vs post



cell-level DE (actually run)

**"treatment changed 120 genes!"  
(true effects: 0)**

simulated data

**one PBMC tutorial pipeline, three datasets, three confident wrong conclusions**

The cold open again: left = decision 2 unanswered, middle = decision 3, right = decisions 1 and 4

# Trap 2: copying parameters across tumours

## What it is

Seeing  $mt < 20\%$  in a GBM paper and copying that number onto your PBMC or a different tumour.



## Why it happens

The number looks like a field convention — but its justification lives in that dataset's distribution.

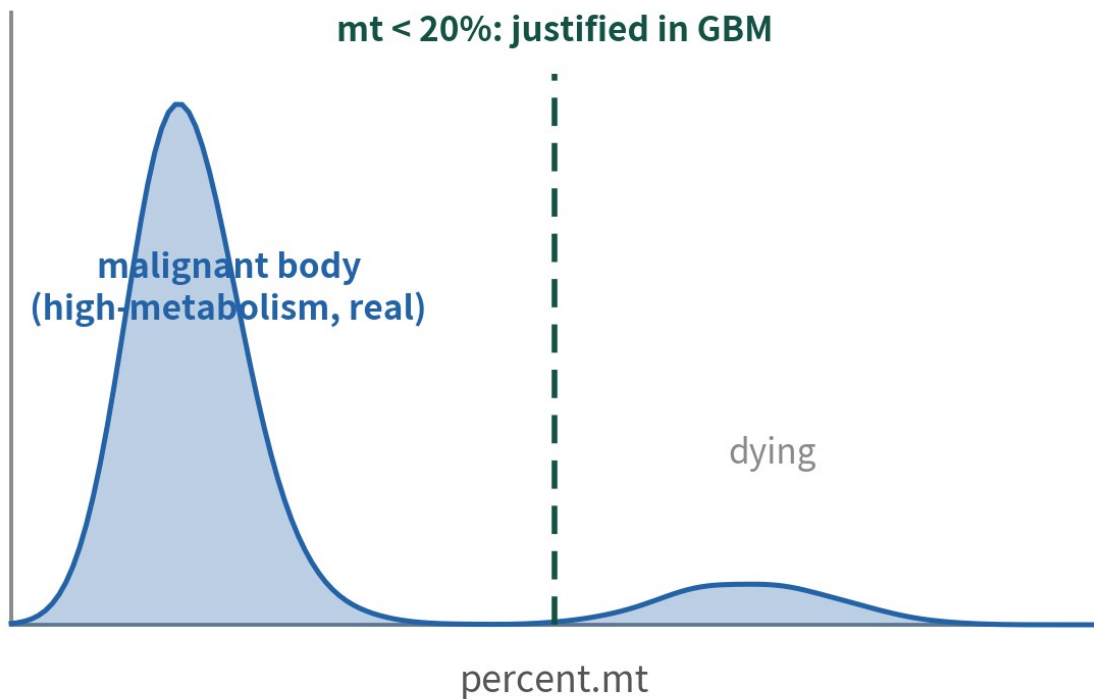
## What it costs you

On PBMC every dying cell between 6–20% sails through; copied the other way (5% onto liver) it massacres healthy cells.

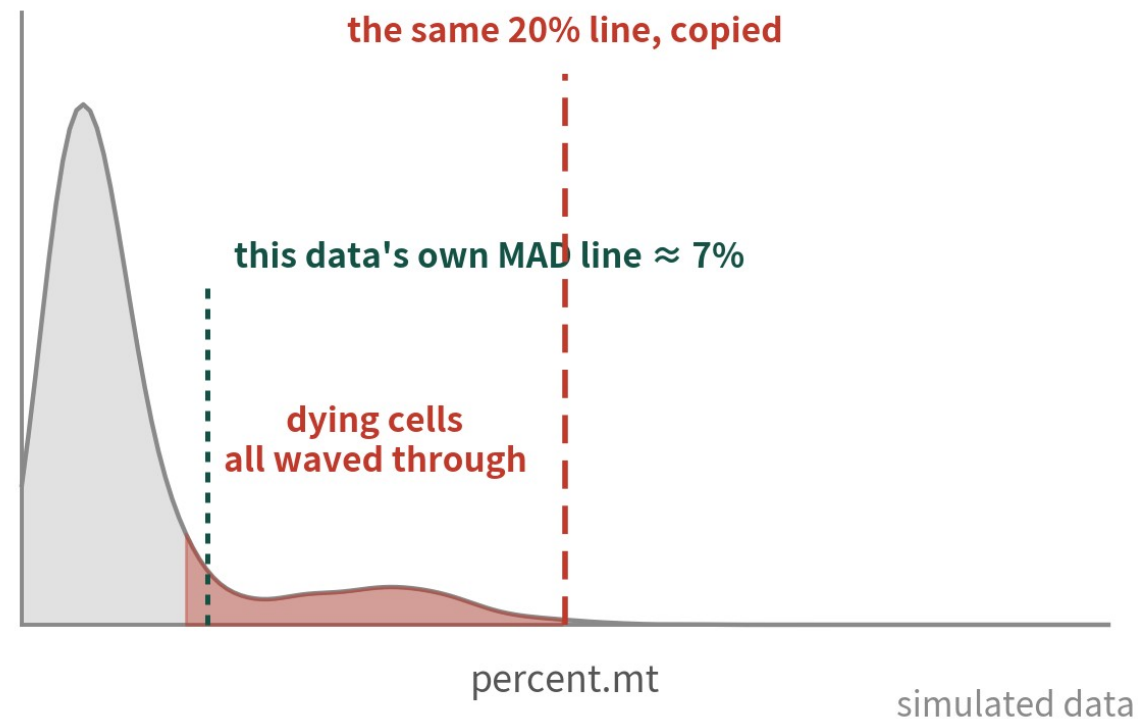


# One number, two fates

GBM: the 20% line was computed



PBMC: the same number becomes a disaster



**a parameter's reason doesn't transfer — only the rule does (inspect, compute MAD, record why)**

What transfers is the rule — inspect, compute MAD, record why — never the number

# Trap 3: ignoring the patient effect

## What it is

Treating cells as independent samples in DE, with the patient level absent from the model.



## Why it happens

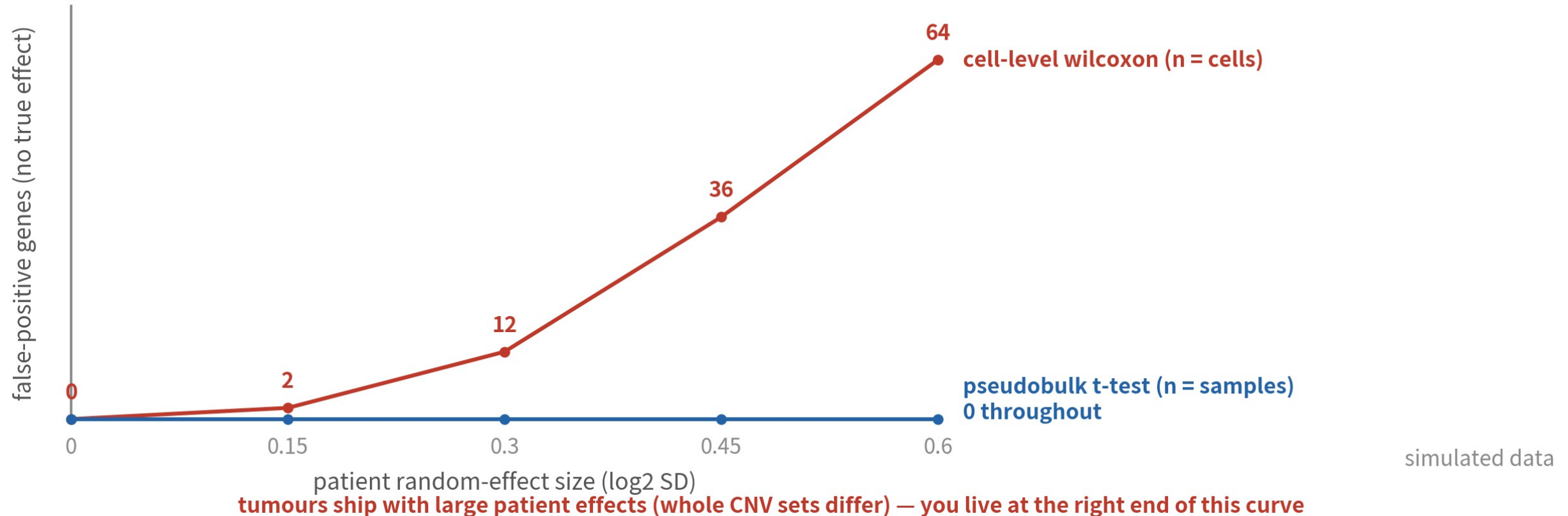
B12's pseudoreplication — but tumours ship with larger patient effects: malignant cells share a whole CNV set.

## What it costs you

Simulated for real: at zero patient effect, zero false positives; at 0.6, 64 of 300 null genes come out 'significant'.

# Patient effect vs false positives, actually run

the larger the patient effect, the worse cell-level FPs (300 null genes, run for real)



Pseudobulk stays at zero throughout — section 3 of R/B19\_strategy.R reproduces this curve

## TRAPS, SUMMED UP

**Complex samples never crash your code.  
They quietly corrupt your conclusions.**

The fix for all three traps is not a new tool — it is answering the five decisions before you start.

# The five-decision quick-reference card

the strategy card: fill all five before touching code — numbers, reasons, records

|                | the question to ask                            | default answer (overridable — write down why)             |
|----------------|------------------------------------------------|-----------------------------------------------------------|
| 1. unit        | who is the independent observation here?       | default: the patient. cells as n = pseudoreplication      |
| 2. integration | which cells should mix, which must stay apart? | align TME; malignant per patient; within-patient pools OK |
| 3. malignancy  | which evidence is available for this cancer?   | big CNV → inferCNV; clean → mutation + phenotype          |
| 4. design      | can batch and biology separate? is it paired?  | pair when you can; mix batches before collection          |
| 5. downstream  | can the data actually carry this analysis?     | check MIN_CELLS / MIN_PAIRS first, then choose            |

Each row: one question, one default. Defaults can be overturned — with a written reason

# Six scenario presets: strategy\_presets()

```
presets <- strategy_presets() # 六場景起點（範例值）  
names(presets)  
presets$AM L$malignancy
```

```
[1] "GBM "      "melanoma"  "breast"  
[4] "liver"     "AML"       "multiregion"  
[1] "分化阻滯 + 突變 / 融合 + 免疫表型; CNV 退居輔助 "
```

Every preset is commented 'example values — adjust to your data': a starting point, never an answer

## ONE-LINE SUMMARY

**'Tumour' is not a singular noun —  
ask what the sample is, then choose the playbook.**

Tools go stale; the five decision points do not.

# Check yourself 1

Across all six scenarios, which of the five decisions never changes?

- A How far integration corrects
- B The patient as the unit of analysis
- C CNV for malignancy calls
- D Trajectories as mandatory downstream

**Answer** B. Integration, malignancy evidence, and downstream routes all change with the tumour — even CNV fails in AML. The one constant: the independent observation is the patient; cells as n is pseudoreplication.



# Check yourself 2

In AML, the 'one patient island per sample' signal often vanishes. Why?

- A Too few AML cells
- B Insufficient sequencing depth
- C Blasts resemble normal progenitors and CNV is often clean
- D Integration corrects blood samples harder

**Answer** C. Patient islands come from large patient-specific CNV dragging whole expression profiles. AML blasts are often near-diploid and stall at the progenitor stage, overlapping normal progenitors — the islands dissolve, and the evidence must change.

# Check yourself 3

**You get melanoma data: 4 pre and 4 post patients, each condition sequenced on a different date. What must you flag for DE?**

- A Use a stricter logFC threshold
- B Treatment and batch are fully confounded — effects can't be attributed; report as exploratory
- C SCTransform will correct it
- D Run DE per batch and intersect

**Answer** B. When condition equals batch, no statistical method can separate them — a design problem, not a correction problem. The real fix happens before collection: pairing, banking, mixed batches. After the fact, honestly downgrade to hypothesis-generating.

NEXT EPISODE

# B18: The ecosystem, and the finale

One tile of the nineteen-episode map remains: what lies beyond this pipeline — multi-omics, spatial, the Python world.

Tools change every year; what is worth carrying with you? See you at the finale.